Mahdi Alam

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PROFILE

MSc Computer Science researcher and Python programmer with six years of Python experience and approximately one year each of React.js and Node.js. Research in self-supervised learning, optimization, reproducible experiments, and NLP retrieval; recent AI-assisted React/TypeScript and Node.js search project.

EDUCATION

Toronto Metropolitan University | Toronto, ON

MSc, Computer Science | Thesis research in artificial intelligence
September 2023 - Expected December 2026 (possible extension to January 2027)

BSc (Honours), Computer Science | September 2019 - June 2023

TECHNICAL SKILLS

Programming: Python, C++, C

Data and machine learning: Pandas, NumPy, SciPy, Scikit-learn, PyTorch, TensorFlow, OpenCV

Methods: Experiment design, quantitative evaluation, ablation studies, clustering, self-supervised learning, optimization

Tools: Git, Docker, Linux

Web development: React.js, Node.js, JavaScript; TypeScript in Document Search Lab

SELECTED SOFTWARE PROJECTS

Document Search Lab | React, TypeScript, Node.js | October 2026

AI-assisted personal project | mahdialam.com/search-lab/ | GitHub source and tests

- Built a React/TypeScript interface and Node.js API for BM25 document ranking, custom in-memory collections, and per-term score explanations.
- Implemented request validation and size limits, accessible form labels, and loading, error, and empty states; verified 15 automated tests, including HTTP integration, plus TypeScript and production build checks.

Sofa Store Website | React, AngularJS, JavaScript, PHP, MySQL | January-May 2022

- Built personalized user accounts and MySQL storage for user and product data; used MVC architecture and Google Maps API integration.

Search Engine - NLP and Retrieval | Python | October-November 2022

- Implemented TF-IDF and BM25 retrieval; trained Naive Bayes and linear SVM baselines and compared retrieval effectiveness across models.

RESEARCH EXPERIENCE

MSc Thesis Research - Self-Supervised Medical Image Registration
Toronto Metropolitan University | Toronto, ON

- Designed a two-phase, coarse-to-fine training strategy combining multi-resolution curriculum learning and adaptive optimization for self-supervised deep learning.
- Improved Dice Similarity Coefficient by up to 1.7 points across public MRI and CT datasets (LPBA40, IXI, OASIS), using quantitative benchmark evaluation.
- Evaluated deformation quality through Jacobian determinant analysis and conducted reproducible Python experiments with ablation studies.

AVAILABILITY

Canadian citizen; authorized to work without sponsorship. Available one month after an offer; willing to relocate within Canada. Expected graduation December 2026, possibly extended to January 2027.